STM Exercise 8

Phase Diagrams

(Fe-C)

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IRON-CARBON (Fe-C) PHASE DIAGRAM

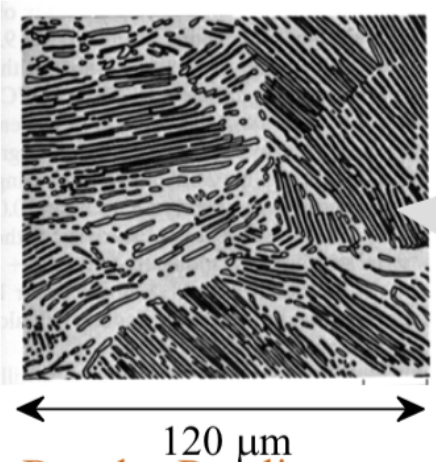
- 2 important points

  -Eutectic (A):

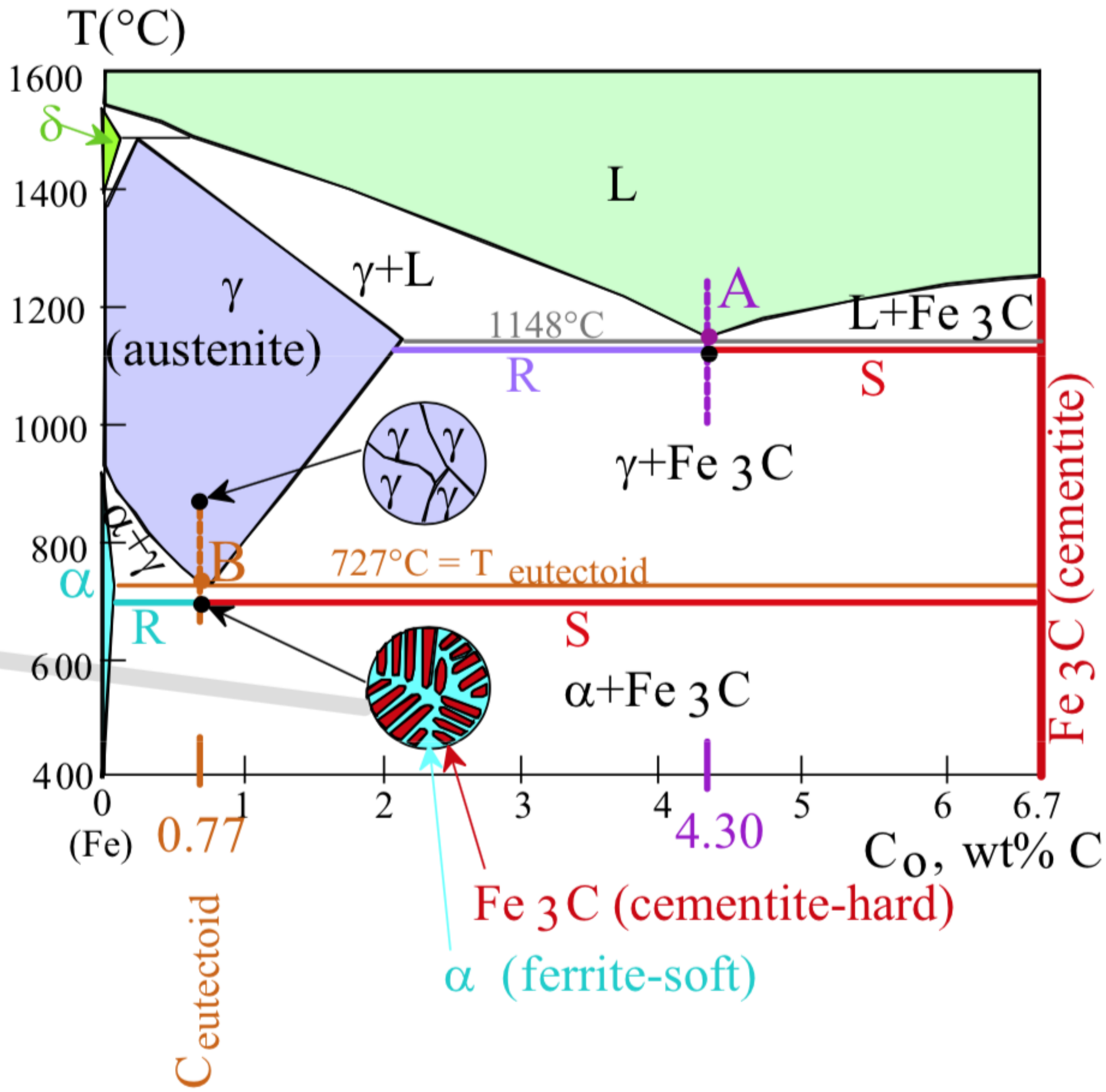
  $L \Rightarrow \gamma + Fe_3C$

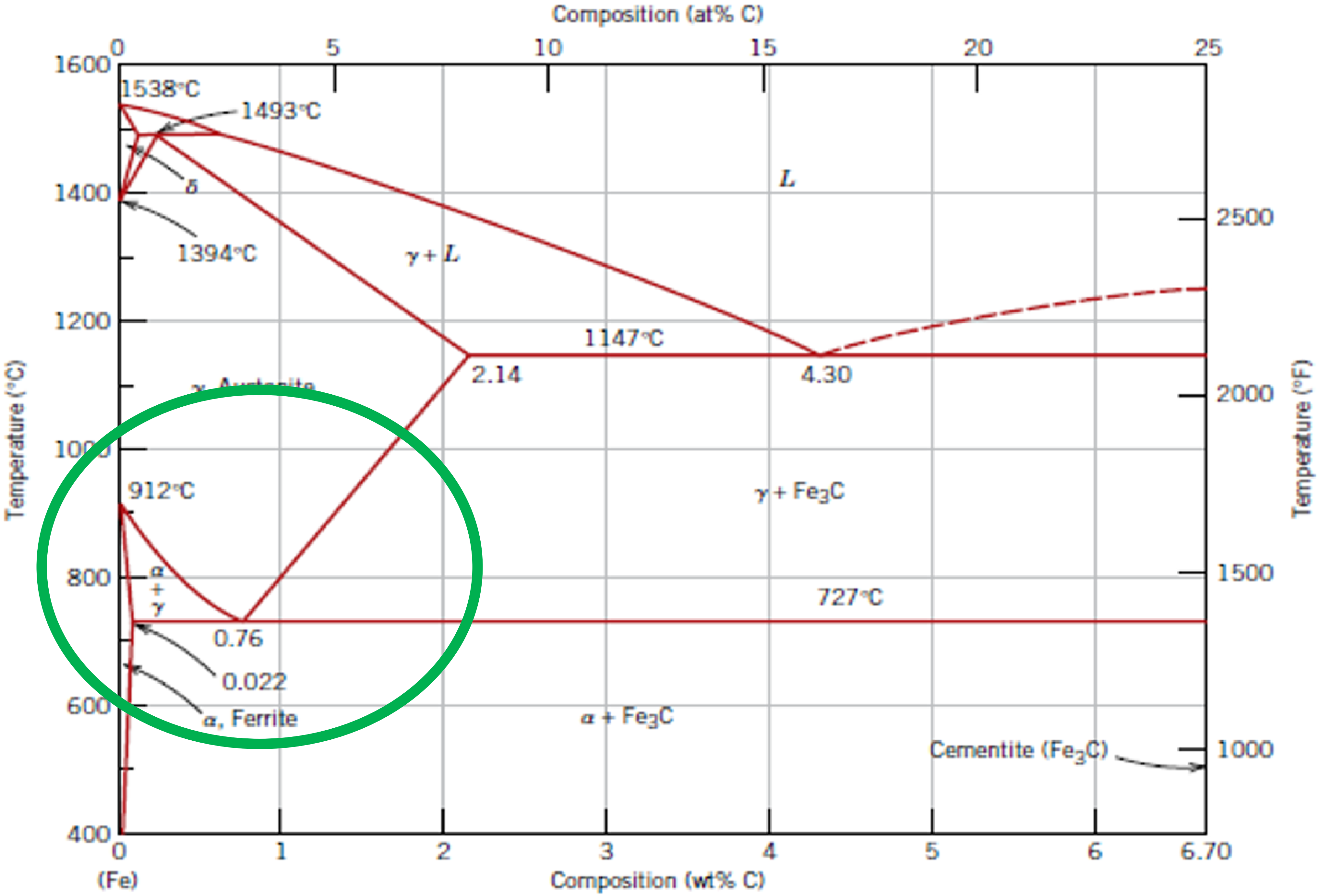
  -Eutectoid (B):

  $\gamma \Rightarrow \alpha + Fe_3C$



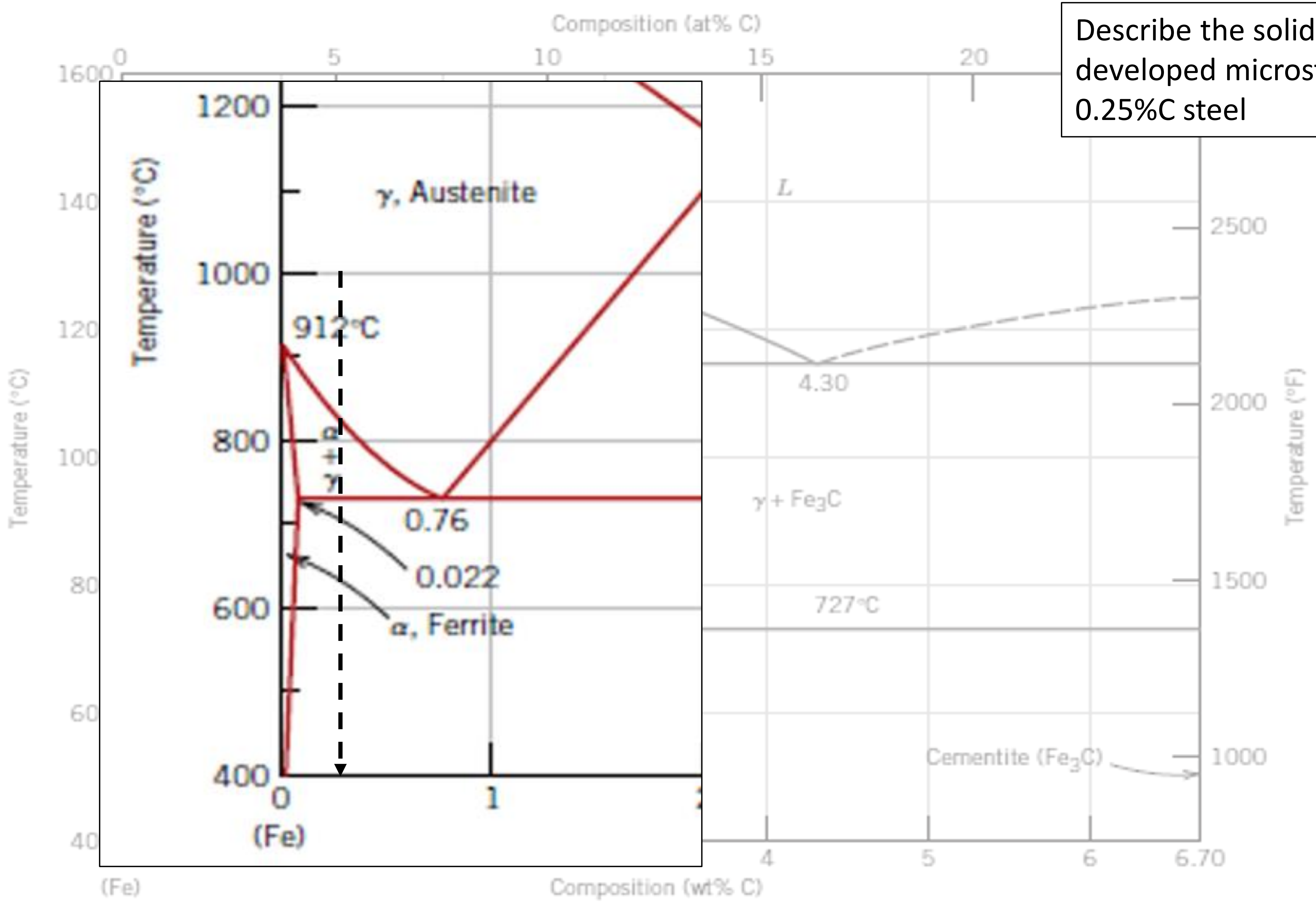
Result: Pearlite = alternating layers of α and Fe_3C phases.





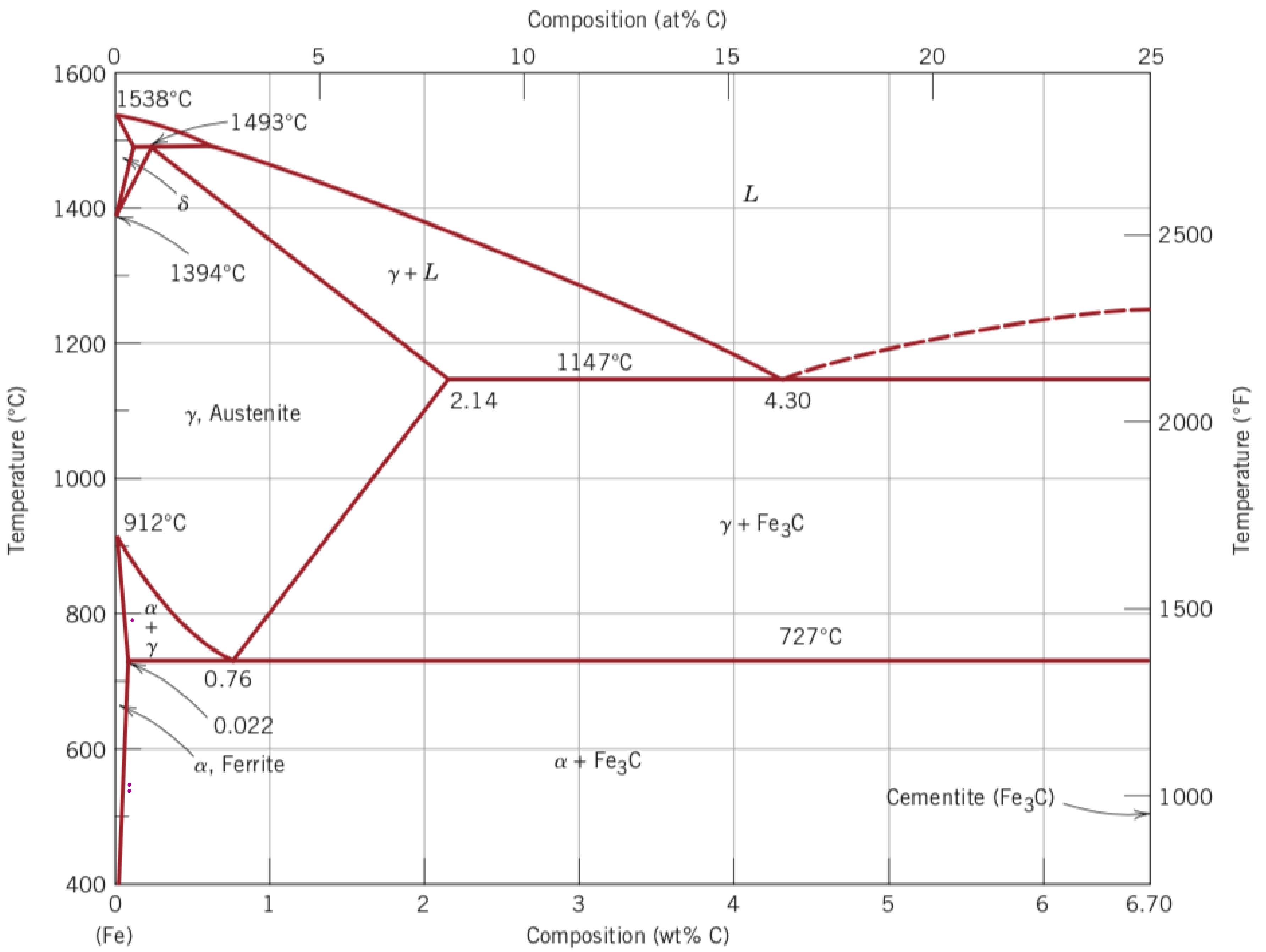
Composition (at% C)
0
5
10
15
20
25
1600
1400
1200
800
600
400
1538°C
1493°C
δ
1394°C
L
γ + L
1147°C
2.14
4.30
2500
2000
1500
1000
Temperature (°C)
Temperature (°F)
912°C
α
+
γ
γ + Fe$_3$C
727°C
0.76
0.022
α, Ferrite
α + Fe$_3$C
Cementite (Fe$_3$C)
0
(Fe)
1
2
3
4
5
6
6.70
Composition (wt% C)

Describe the solidification (evidencing developed microstructures) of a 0.25%C steel



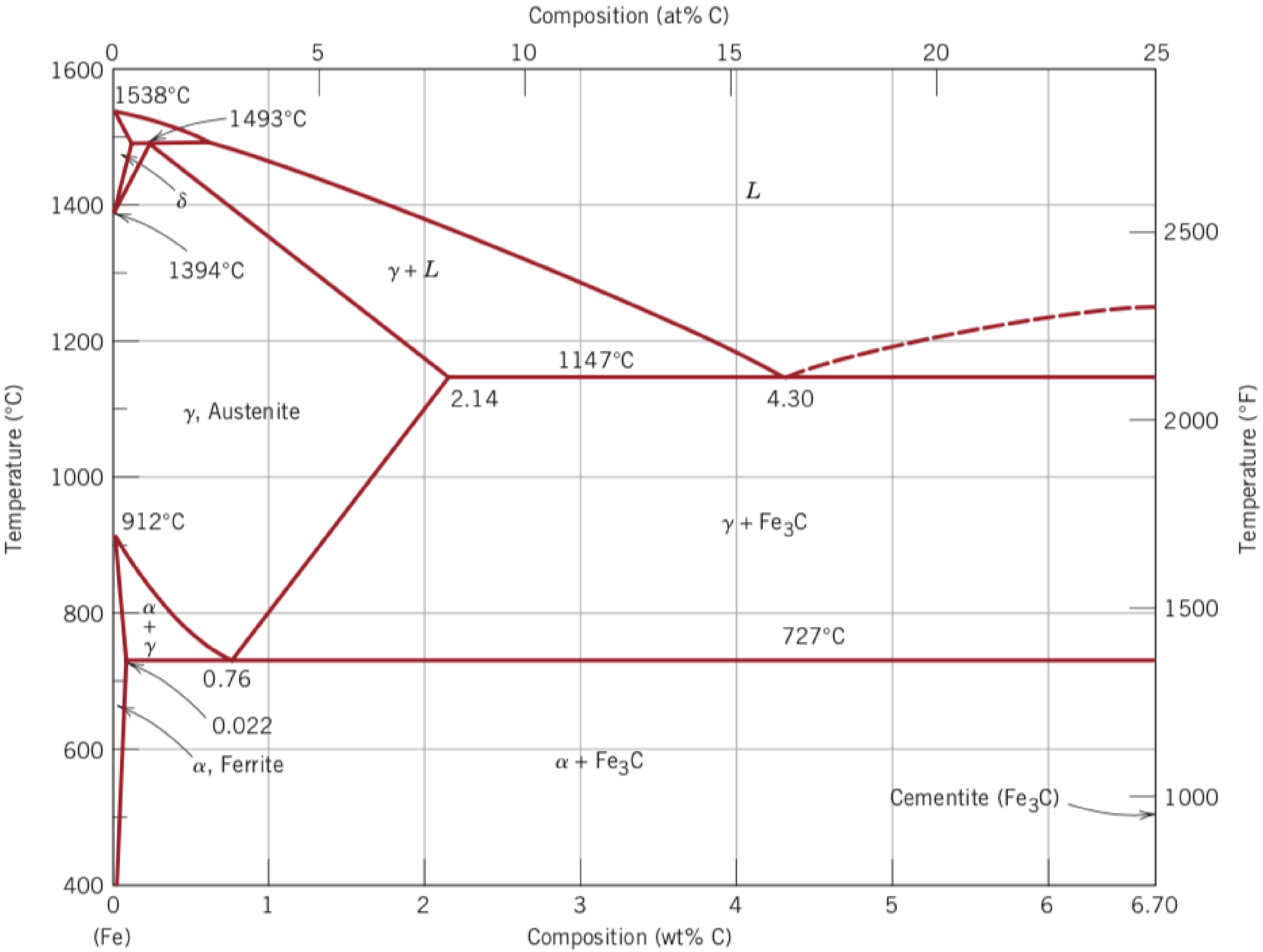
Use the Fe-Fe_3C phase diagram to calculate:

-the percentage of primary (pro-eutectoid) ferrite for a 0,2 % C steel



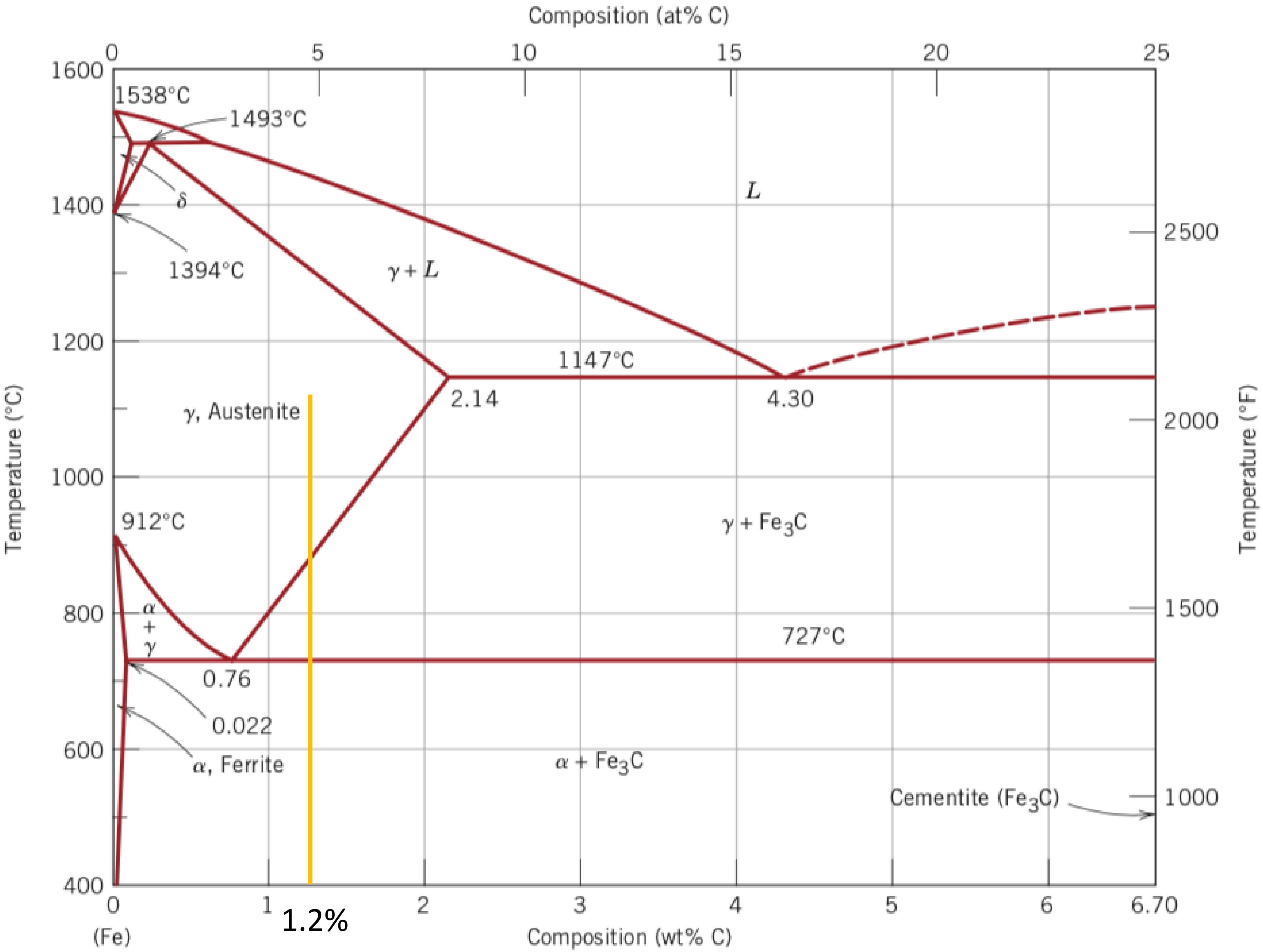
Use the Fe-Fe3C phase diagram to calculate:

-the percentage of pearlite for a 0,2 % C steel



Use the Fe-Fe3C phase diagram to calculate:

-the percentage of pearlite for a 1,2 % C steel



Use the Fe-Fe3C phase diagram to calculate:

-the percentage of primary (pro-eutectoid) cementite for a 1,2 % C steel